Database project

Dr/Mohamed Hasan

Project: Travel Agency Database

Group: 7

Team members:

Sunday: 12:00

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Introduction:

A travel agency database is designed to efficiently store, organize, and manage all the information related to customers, trips, bookings, hotels, employees, payments, and reviews.

The main goal of this database is to support the agency's operations by providing an integrated system that tracks customer reservations, trip destinations, hotel arrangements, and payment details.

This database helps the agency improve its services by maintaining accurate records of each transaction, ensuring smooth communication between customers and travel agents, and simplifying the management of travel packages and hotel partnerships.

The database includes several entities such as Customer, Trip, Booking, Employee, Hotel, Payment, and Review. Each entity contains important attributes and relationships that reflect real-world interactions within the travel business. For example, a customer can book multiple trips, each trip may include hotel information and payment records, and customers can provide reviews after their trips.

By using this database system, the travel agency can achieve better data consistency, reduce redundancy, and generate reports that support decision-making, customer satisfaction, and business growth.

Entities and their attributes:

1-Customer

- Customer_id (PK)
- First_name
- Last_name
- Phone_number
- Email_address

2- Trip

- Trip_id (PK)
- Trip_Name
- Destination
- Start_Date
- End_Date

3- Booking

- Booking_id (PK)
- Booking_date
- Status (confirmed, pending, cancelled)
- Customer_id (FK)
- Employee_id (FK)
- Trip_id (FK)
- Hotel_id (FK)

4-Employee

- Employee_id (PK)
- First_name
- Last_name
- Email
- Phone_number
- Salary

5- Hotel

- Hotel_id (PK)
- Name
- City
- Rating

6- Payment

- Payment_id (PK)
- Method
- Amount
- Payment_date
- Booking_id (FK)

7-Review

- Review_id (PK)
- Review_date
- Rating
- Comment
- Booking_id (FK)

Database Schema

Customer Table

Customer_id (PK)
First_name
Last_name
Phone_number
Email_address

Trip Table

Trip_id (PK)
Trip_Name
Destination
Start_Date
End_Date

Booking Table

Booking_id (PK)
Booking_date
Status (confirmed, pending, cancelled)
Customer_id (FK)
Employee_id(FK)
Trip_id (FK)
Hotel_id (FK)

Employee

Employee_id(PK)
First_name
Last_name
Email
Phone_number
Salary

Hotel Table

Hotel_id (PK)
Name
City
Rating

Payment Table

Payment_id (PK)
Method
Amount
Payment_date
Booking_id (FK)

Review Table

Review_id (PK)
Review_date
Rating
Comment
Booking_id (FK)

Database Instance:

Customer

Customer_id	First_name	Last_name	Phone	Email
1	Mohamed	Aly	01123456789	mohamed@gmail.com
2	Ahmed	Ashraf	01234565434	ahmed@gmail.com
3	Mona	Zeyad	01087654674	mona@gmail.com

Trip

Trip_id	Trip_Name	Destination	Start_Date	End_Date
T1	Trip 663	Tokyo	15-07-2025	25-07-2025
T2	Trip 504	Paris	01-06-2025	10-06-2025
T3	Trip 729	Roma	20-08-2025	28-08-2025

Booking

Booking_id	Booking _date	Status	Customer_id	Employee_id	Trip_id	Hotel_id
B1	10-05-2025	Confirmed	1	A1	T2	H2
B2	02-07-2025	Pending	2	A2	T1	H1
B3	12-08-2025	Confirmed	3	A3	T3	H3

Employee

Employee_id	First_name	Last_name	Phone	Email	Salary
A1	Sara	Aly	01123456789	Sara@gmail.com	9,000
A2	Youssef	Ashraf	01234565434	youssef@gmail.com	8,000
A3	Lena	Saied	01087654674	lena@gmail.com	7,000

Hotel

Hotel_id	Name	City	Rating
H1	Sakura Hotel	Tokyo	4.5
H2	Eiffel Stay	Paris	4.8
H3	Roma Palace	Roma	4.2

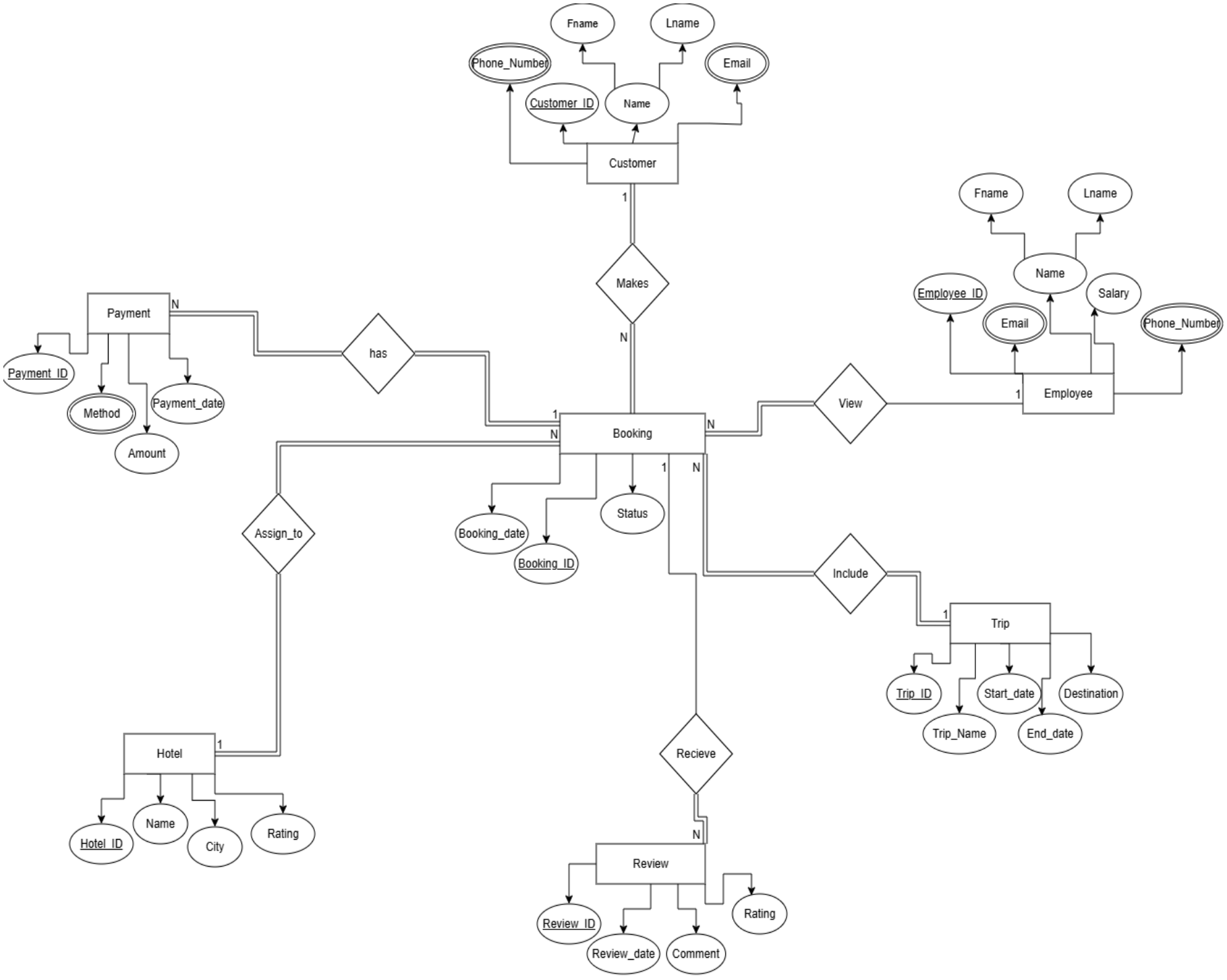
Payment

Payment_id	Method	Amount	Payment_date	Booking_id
P1	Credit Card	9,000	10-05-2025	B1
P2	Cash	7,500	14-07-2025	B2
P3	Bank Transfer	8,200	12-08-2025	B3

Review

Review_id	Review_date	Rating	Comment	Booking_id
R1	11-06-2025	4.5	Nice	B1
R2	27-07-2025	4.8	A good place	B2
R3	29-08-2025	4.2	Hotel need some fixes	B3

ERD


Fname
Lname
Phone_Number
Email
Customer_ID
Name
Customer
1
Makes
N
Payment
N
has
Payment_ID
Method
Payment_date
Amount
1
N
Booking
N
View
Fname
Lname
Name
Employee_ID
Email
Salary
Phone_Number
1
Employee
Assign_to
Booking_date
Booking_ID
Status
1
N
Include
1
Trip
Trip_ID
Start_date
Destination
Trip_Name
End_date
1
Hotel
Hotel_ID
Name
City
Rating
Recieve
N
Review
Review_ID
Review_date
Comment
Rating

Mapping

1- Employee

Emp_id	E-mail	First-name	Last-name	Salary

Employee phone_number

Emp_id	Emp_number

2- Trip

Trip_id	Trip_name	Destination	Start_date	End_Date

3- Customer

Customer_id	First_name	Last_name	Email

Customer phone_number

Customer_id	Customer_phone

4- Booking

Booking_id	Booking_date	Confirm	Cancel	Pending	Customer_id	Employee_id
Trip_id	Hotel_id					

4- Booking

Booking_id	Booking_date	Confirm	Cancel	Pending	Customer_id	Employee_id
Trip_id	Hotel_id					

5- Hotel

Hotel_id	Name	City	Rating

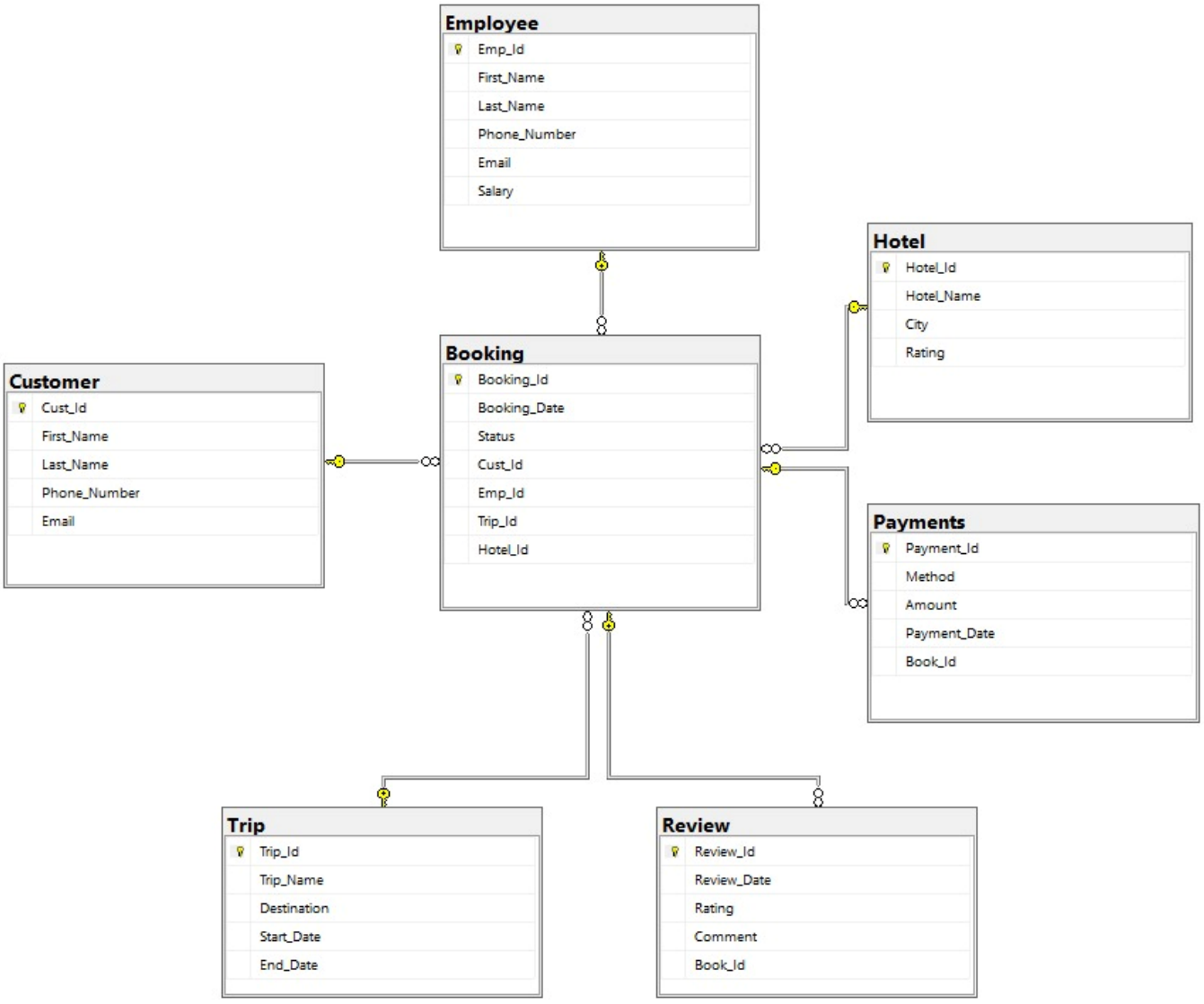
6- Payment

Payment_id	Method	Amount	Payment_date	Booking_id

7-Review

Review_id	Review_data	Rating	Comment	Booking_id

SQL


Employee
Emp_Id
First_Name
Last_Name
Phone_Number
Email
Salary
Hotel
Hotel_Id
Hotel_Name
City
Rating
Customer
Cust_Id
First_Name
Last_Name
Phone_Number
Email
Booking
Booking_Id
Booking_Date
Status
Cust_Id
Emp_Id
Trip_Id
Hotel_Id
Payments
Payment_Id
Method
Amount
Payment_Date
Book_Id
Trip
Trip_Id
Trip_Name
Destination
Start_Date
End_Date
Review
Review_Id
Review_Date
Rating
Comment
Book_Id